UNIVERSITY OF ILLINOIS CHICAGO

Center for the Advancement of Teaching Excellence

Erin Stapleton-Corcoran, *CATE Instructional Designer*
January 25, 2023

Bloom's Taxonomy of Educational Objectives

WHAT?

Bloom's taxonomy is a hierarchical model used for classifying learning objectives by levels of complexity and specificity. Bloom's Taxonomy was created to outline and clarify how learners acquire new knowledge and skills. Though the original intention of the taxonomy was to serve as an assessment tool, Bloom's taxonomy is effective in helping instructors identify clear learning objectives as well as create purposeful learning activities and instructional materials.

Bloom's taxonomy emerged from a 1948 meeting of university educators – chaired by Benjamin Bloom – who brainstormed a theoretical model of learning that identified educational objectives to aid in the creation of testing items. The committee later expanded its initial framework to three learning domains:

- Cognitive: knowledge or thinking
- Affective: growth in feelings or emotional areas (attitude or self)
- Psychomotor: manual or physical skills



Bloom's Taxonomy consists of three learning domains: cognitive, affective and psychomotor.

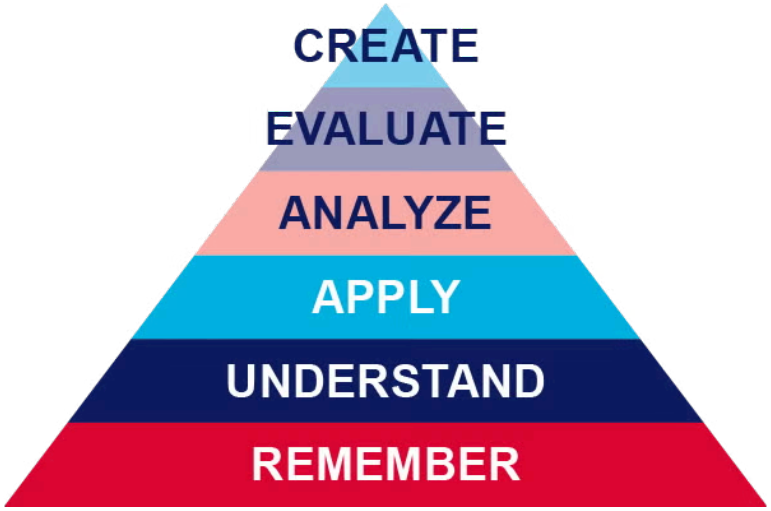
Cognitive Domain

The cognitive domain went through numerous revisions before a finalized version was published (Bloom 1956).

The cognitive domain has been the primary focus in education and has become shorthand for Bloom's Taxonomy as a result. The cognitive domain is made up of six levels of objectives. These levels are organized by hierarchy, moving from foundational skills to higher-order thinking skills.

In 2001 Anderson and Krathwohl revised Bloom's levels from nouns to verbs, and this is the version of the taxonomy used today.

- Remember: retrieve relevant knowledge from memory.
- Understand: determine the meaning of instructional messages.
- Apply: use a procedure in a given situation.
- Analyze: break materials into components and determine how they work together.
- Evaluate: make judgments based on criteria and standards.
- Create: create a new or original work.



Anderson and Krathwohl revised Bloom's levels from nouns to verbs.



Anderson and Krathwohl's Two-Dimensional Taxonomy

When revising Bloom's Taxonomy in 2001, Anderson and Krathwohl also added the knowledge dimension to the taxonomy. The knowledge dimension consists of four dimensions, which are:

- Factual knowledge (basic elements to learn or solve problems in the discipline)
- Conceptual knowledge (interrelationships between basic elements within a larger context)
- Procedural knowledge (methods in the discipline)
- Metacognitive knowledge (awareness of how learning work in relation to one's self)

Based on this two-dimensional taxonomy, Anderson and Krathwohl developed a matrix for combining cognitive processes and knowledge dimensions which is shown to the left.

See the "How [https://teaching.uic.edu/blooms-taxonomy-of-educational-objectives/#how]
" section of this teaching guide to learn more about using the cognitive domain to craft learning objectives, assessments, and instructional materials and learning activities for your courses as well as how to implement the cognitive domain/knowledge dimension matrix when using Bloom's Taxonomy in the classroom.

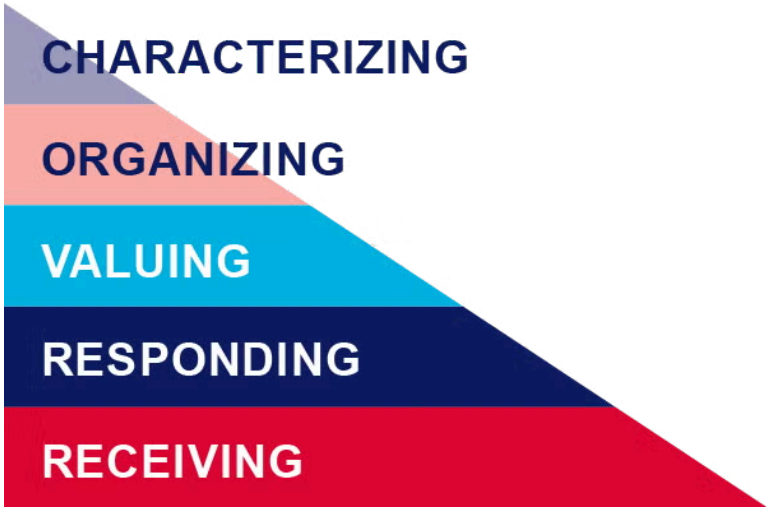
Affective Domain

The affective domain was first published in 1964 (Krathwohl et al, 1964). The affective domain outlines skills and behaviors that correspond to attitudes and values and as the learner progresses through the levels of the affective domain, they become self-reliant and internally motivated. Learning objectives aligned to the affective domain tend to be the hardest to articulate initially and often appear difficult to assess at first glance. However, affective outcomes often represent the outcomes most closely related to deeper thinking and lifelong learning.

The affective domain contains five levels, from lowest to highest:

- Receiving: Willing to listen and receive knowledge.
- Responding: Actively participates and engages in knowledge transfer.
- Valuing: Finds value and worth in one's learning with motivation to continue.
- Organizing: Integrates and compares values, resolves conflict between these values, and orders them according to priorities.
- Characterizing: Creates a value system that controls behavior. The behavior is pervasive, consistent, predictable, and characteristic of the learner.

See the "How [https://teaching.uic.edu/resources/teaching-guides/learning-principles-and-frameworks/blooms-taxonomy-of-educational-objectives/#how]
" section of this teaching guide to learn more about using the affective domain to craft learning objectives, assessments, and instructional materials and learning activities for your courses.



(Krathwohl et al., 1964)

Psychomotor Domain

Bloom and his colleagues did not create subcategories for skills in the psychomotor domain, but other educators did (Simpson 1966, 1972; Dave, 1970; Harrow, 1972). The psychomotor domain includes physical movement, coordination, and motor skills. Development of these skills requires practice and is measured in terms of speed, precision, distance, procedures, or technical execution. For the purpose of this teaching guide, we will explore Simpson's version of the psychomotor domain, which has the following seven levels:

- **Perception**: Use sensory cues to guide actions or movements.
- **Set**: Demonstrates a readiness (physically, mentally, emotionally, and spiritually) to take action to perform the task or objective. (NOTE: This level of the Psychomotor domain is closely related to the "Responding to phenomena" level of the Affective domain).

- **Guided response**: Knows steps required to complete the task or objective and learns through trial and error by practicing.
- **Mechanism**: Performs task or objective in a somewhat confident, proficient, and habitual manner.
- **Complex overt response**: Performs task or objective in a confident, proficient, and habitual manner. Expert level, high proficiency and performs with accuracy.
- **Adaptation**: Performs task or objective and can modify actions to account for new or problematic situations.
- **Origination**: Create new procedures and solutions to approach various situations.



Simpson's seven levels of the psychomotor domain.

See the "How [https://teaching.uic.edu/blooms-taxonomy-of-educational-objectives/#how]" section of this teaching guide to learn more about using the psychomotor domain to craft learning objectives, assessments, and instructional materials and learning activities for your courses.

WHY?

There are several benefits to applying Bloom's taxonomy to your teaching.

- Bloom's Taxonomy can help instructors craft clear, actionable learning objectives. Clear, actionable learning objectives help students understand the skills and knowledge they will gain during the course.
- Bloom's Taxonomy can help instructors appropriately align instruction to the learning objectives, including the planning of learning activities and the delivery of instructional materials (Raths 2002).
- Bloom's taxonomy helps instructors create valid and reliable assessments by aligning course learning objectives to any given level of student understanding or proficiency. Crooks (1998) suggests that much of college assessment involves recalling memorized facts, which only addresses the first level of learning. However, Bloom's Taxonomy aids instructors in creating assessments that address all six levels of the cognitive domain.
- Bloom's Taxonomy has been shown to enhance students' higher-order thinking skills, such as critical thinking. Bissell and Lemons (2006) used Bloom's Taxonomy to assess critical-thinking skills in an introductory biology course. They developed a process by which they prepared questions with both content and critical-thinking skills in mind, and prepared grading rubrics that specified how to evaluate both the content and critical-thinking aspects of an answer. Using this methodology helped Bissell and Lemons clarify the course goals (for instructors and students), improve student metacognition, and expose student misconceptions about the course content.

HOW?

How can you use Bloom's Taxonomy to craft learning objectives?

Bloom's Taxonomy can help you write clear learning objectives [https://teaching.uic.edu/learning-objectives/], which are a description of what the learner must be able to do upon completion of an educational activity. A well-written learning objective outlines the knowledge, skills, and/or attitude the learners will gain from the educational activity and does so in an observable and measurable way.

More specifically, Bloom's can help you identify the level, criteria, or standards for the knowledge, skills, abilities, competencies, attitudes, or values that your students are expected to be able to demonstrate.

For the cognitive, affective, and psychomotor domains, there are action verbs that align with each level. These action verbs are invaluable in crafting effective learning objectives.

Use the following steps to craft learning objectives using Bloom's taxonomy:

1. Select the domain (cognitive, affective, or psychomotor) for the learning objective you want to design.
2. Determine at what complexity level you want students to demonstrate their learning.
3. Select an action verb that is aligned to the domain level that you want students to demonstrate.

Consult the tables below to view action verbs that align with each level of the cognitive, affective, and psychomotor domains, as well as examples for each of the levels.

Cognitive domain

This domain is focused on intellectual skills such as critical thinking, problem-solving, and creating a knowledge base. The cognitive hierarchy consists of six levels, which span from simple memorization designed to build the knowledge of learners, to creating something new based on previously-learned information.

REMEMBER

UNDERSTAND

APPLY

ANALYZE

EVALUATE

CREATE

Examples of Learning Objectives in the Cognitive Domain across the disciplines

LINGUISTICS

BIOLOGY

ENGLISH / LITERATURE

PUBLIC HEALTH

ENGINEERING

Affective domain
The affective domain focuses on a student's attitudes, values, and interests. Composed of five levels, this domain begins with receiving and listening to information, and extends to characterization, or internalizing values and consistently acting upon these values.

RECEIVING

RESPONDING

VALUING

ORGANIZING

CHARACTERIZING

Examples of Affective Learning Objectives Across the Disciplines

ENGLISH

ENGINEERING

SOCIOLOGY

Psychomotor domain
The psychomotor domain focuses on a student's ability to physically accomplish tasks and to perform nonverbal communication and expressive activities. The psychomotor domain consists of seven levels.

PERCEPTION

GUIDED RESPONSE

MECHANISM

COMPLEX OVERT RESPONSE

ADAPTATION

ORIGINATION

How can you use Bloom's Taxonomy in the classroom?
There are many ways that Bloom's can be applied to course planning to ensure effective instruction and student learning is occurring. Several approaches to lesson planning that build on Bloom's Taxonomy are described below.

Expertise pathways

Expertise pathways involve using increasingly processes to accumulate deeper understandi understanding or expertise can be accomplis Bloom's levels to explore a discrete learning t below demonstrates expertise in the cognitive particular play script (Gershon 2013).

Remember the names and relationships of a cast of characters in a play	A multip test the
Explain the main ideas of a play	Write a s summar importa
Apply the main ideas/themes in the play to another context	Write an one of th
Analyze the roles of each character in the play and their relationships to each other.	Write an the anta the play.
Evaluate the decisions of characters in the play, and support your evaluation with textual evidence.	Write a r in the pla rejecting evidence persona consequ
Create a new and unique piece of writing using similar plot devices.	Create a devices



Differentiated instruction (all/most/some)

Three-part lesson

Teach with Anderson and Krathwohl's Two Dimension Taxonomy: Cognitive Processes and Knowledge Dimensions

HOW TO USE/CITE THIS GUIDE



CC BY-NC 4.0
Creative Commons Attribution-NonCommercial 4.0 International

- This work is licensed under Creative Commons Attribution-NonCommercial 4.0 International.
- This license requires that reusers give credit to the creator. It allows reusers to distribute, remix, adapt, and build upon the material in any medium or format, for noncommercial purposes only.

Please use the following citation to cite this guide:
Stapleton-Corcoran, E. (2023). “Bloom’s Taxonomy of Educational Objectives.” Center for the Advancement of Teaching Excellence at the University of Illinois Chicago. Retrieved [today’s date] from https://teaching.uic.edu/blooms-taxonomy-of-educational-objectives/ [https://teaching.uic.edu/blooms-taxonomy-of-educational-objectives/]

CC BY-NC 4.0 Deed [https://creativecommons.org/licenses/lnc/4.0/]

ADDITIONAL RESOURCES

ADDITIONAL RESOURCES

REFERENCES

REFERENCES